

Equivalent Fractions

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Equivalent fractions name the **same amount** in different ways. To make one, **multiply** the top and bottom by the **same number**, or **divide** the top and bottom by the same number. You must do the same thing to both. Example: $1/2 \times 2/2 = 2/4$, and $2/4 \times 2/2 = 4/8$, so **$1/2 = 2/4 = 4/8$** .

Round 1 — Retrieve It

1. What do equivalent fractions have in common?

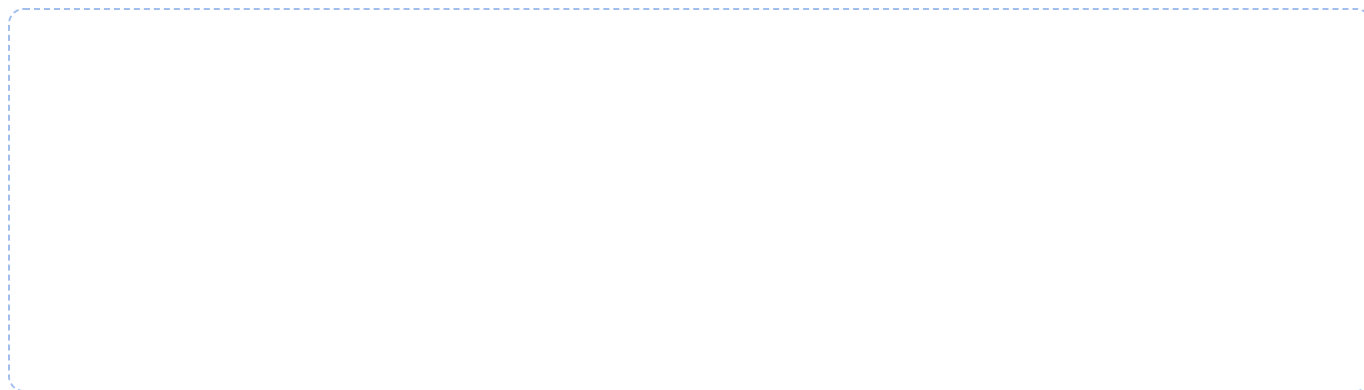
2. To make an equivalent fraction, what do you do to the top and bottom?

3. Write two fractions equal to $1/2$.

4. Besides multiplying, what other operation makes an equivalent fraction?

Round 2 — Sketch It

Draw two same-size bars. Split one into halves and shade $1/2$. Split the other into fourths and shade $2/4$. Show they cover the same amount.



Equivalent Fractions — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Make a fraction equal to $\frac{2}{3}$ by multiplying the top and bottom by 2. Show your work.

2. Write $\frac{6}{8}$ in simplest form by dividing the top and bottom by the same number.

3. Fill in the blank: $\frac{3}{4} = \frac{\quad}{8}$. What did you multiply by?

Round 4 — Reason from Evidence

Jay says $\frac{1}{2} = \frac{2}{3}$ because he added 1 to the top and 1 to the bottom. Is he correct? Explain the right way to make an equivalent fraction.

Equivalent Fractions — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how to make an equivalent fraction. Use the words multiply, divide, and same number.

C**Connect**

Connect equivalent fractions to measuring. Tell how $\frac{1}{2}$ cup and $\frac{2}{4}$ cup give the same amount.

E**Extend**

Start with $\frac{1}{3}$ and write three fractions equal to it. Show what you multiplied by each time.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Equivalent Fractions

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. They name the same amount (the same value).
2. Multiply (or divide) both by the same number.
3. Examples: $\frac{2}{4}$, $\frac{3}{6}$, $\frac{4}{8}$ (any correct pair).
4. Dividing the top and bottom by the same number.

Sketch It — look for

Two same-size bars: one halved with 1 part shaded ($\frac{1}{2}$), one in fourths with 2 parts shaded ($\frac{2}{4}$). Shaded areas match.

Apply It

1. $\frac{2}{3} \times \frac{2}{2} = \frac{4}{6}$.
2. $\frac{6}{8} \div \frac{2}{2} = \frac{3}{4}$.
3. $\frac{3}{4} = \frac{6}{8}$. Multiply top and bottom by 2.

Reason from Evidence — look for

No. You cannot add the same number; you must multiply or divide top and bottom by the same number. $\frac{1}{2} = \frac{2}{4}$, not $\frac{2}{3}$.

ACE — Extend look-for

Three fractions equal to $\frac{1}{3}$, such as $\frac{2}{6}$, $\frac{3}{9}$, $\frac{4}{12}$, each showing multiplication by $\frac{2}{2}$, $\frac{3}{3}$, $\frac{4}{4}$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.